Beyond RGB 2.5

Under the Hood

COMPUTATIONAL PIPELINE: OVERVIEW

PRE-PROCESSING

- Read in RAW images
- Bit-depth scaling
- Dark current correction
- Flat-fielding
- Registration

PROCESSING

- Extract camera signals from color target patches
- Colorimetric calibration
- Spectral reflectance calibration
- Optional: verification of calibrations

RENDERING & OUTPUT

- Calculate color-calibrated images and convert to ProPhoto TIFF
- Save and export calibration files and images
- Interactively estimate spectral reflectance

1. PRE-PROCESSING

Input image format:

Linear RAW camera data captured in some proprietary format according to the camera manufacturer (e.g. .CR2, .NEF, .ARW…). These image data are provided as-is, or losslessly converted prior to processing and provided as linear RAW TIFFs.

Input image set:

Each image pair contains **2 RGB images** that were captured under two different conditions. These are combined into a single image stack that has **6 spectral channels**.

The two different conditions may be created by changing the lighting or applying filters to either the lights or the camera.

An RGB with 3 channels

Another RGB with 3 channels

Which are combined to give us our 6-channel spectral image stack ☺

Input image set: Single image processing

Each input image set must contain **at least four pairs** of RGB images

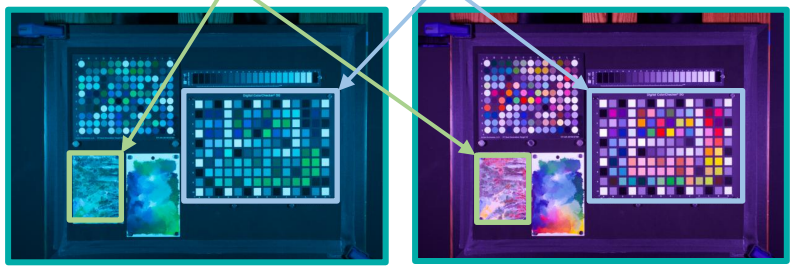
1. Dark current image pair
2. Flatfield image pair
3. Target image pair
4. Object image pair

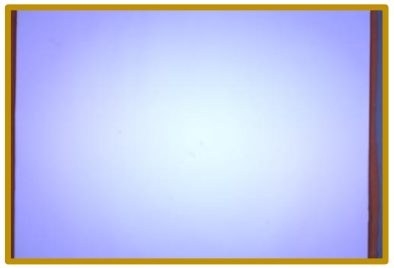
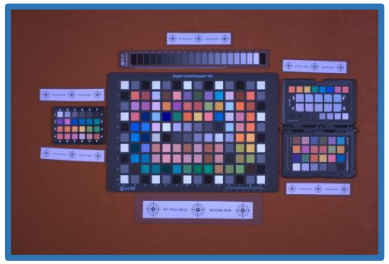
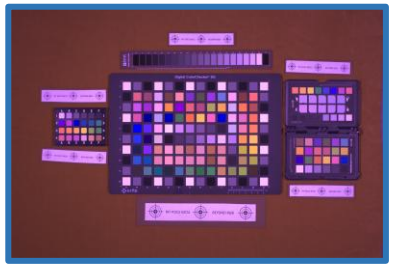
Notes:

If only three pairs of images are uploaded, BeyondRGB assumes that the target and the object are located in the same image. For example:

Object and Target image pair

3 & 4.

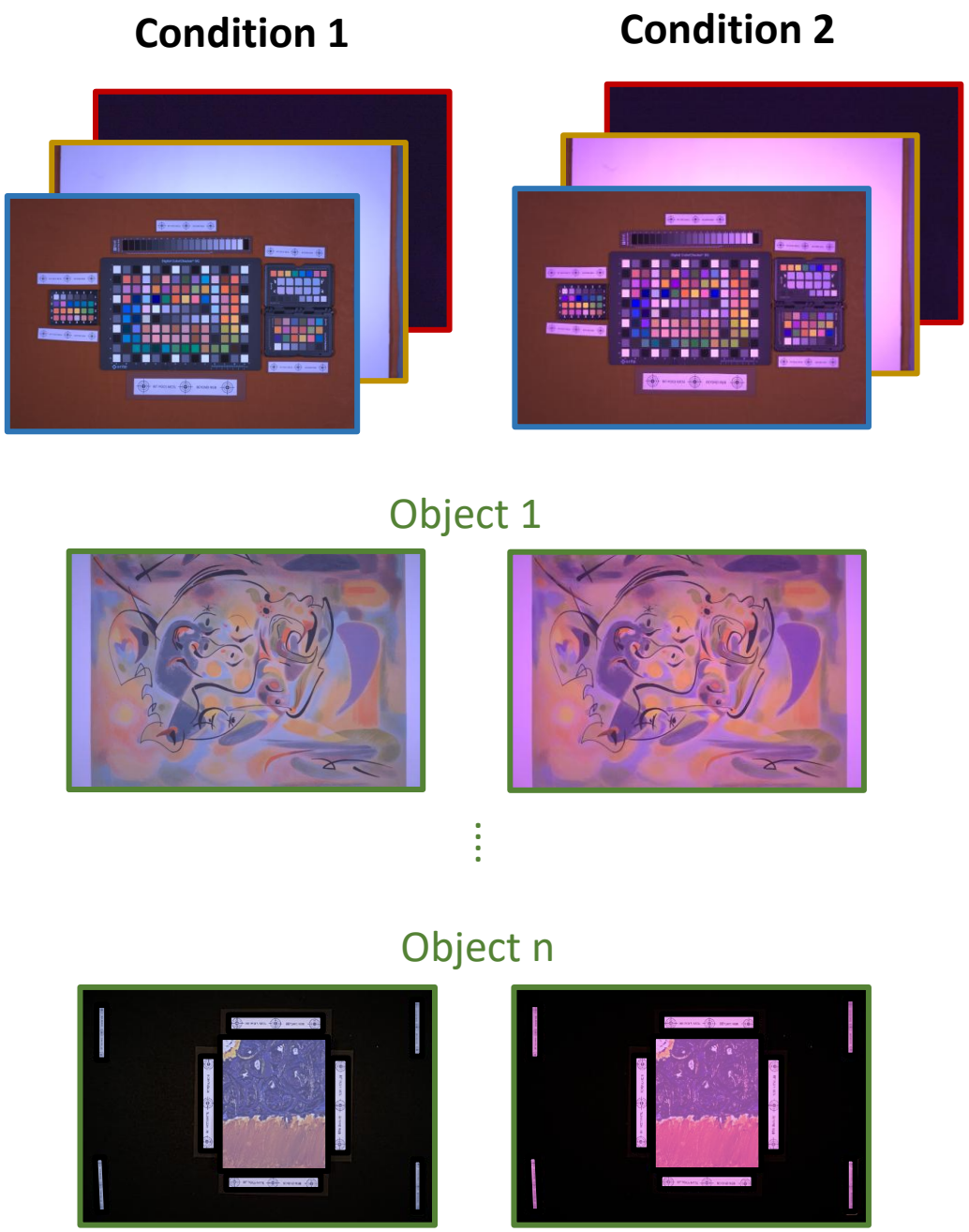


	Condition 1	Condition 2
	Dark current image pair	
1.		
	Flatfield image pair	
2.		
	Target image pair	
3.		
	Object image pair	
4.		

Input image set: Batch processing

If there are multiple objects captured at the same time, all of the **object image** pairs can be processed simultaneously.

Only one set of **dark current**, **flatfield**, and **target** image pairs is required.



Bit depth scaling:

The RAW camera data is usually captured by the sensor at either 12-bit (4095) or 14-bit (16383) depth.

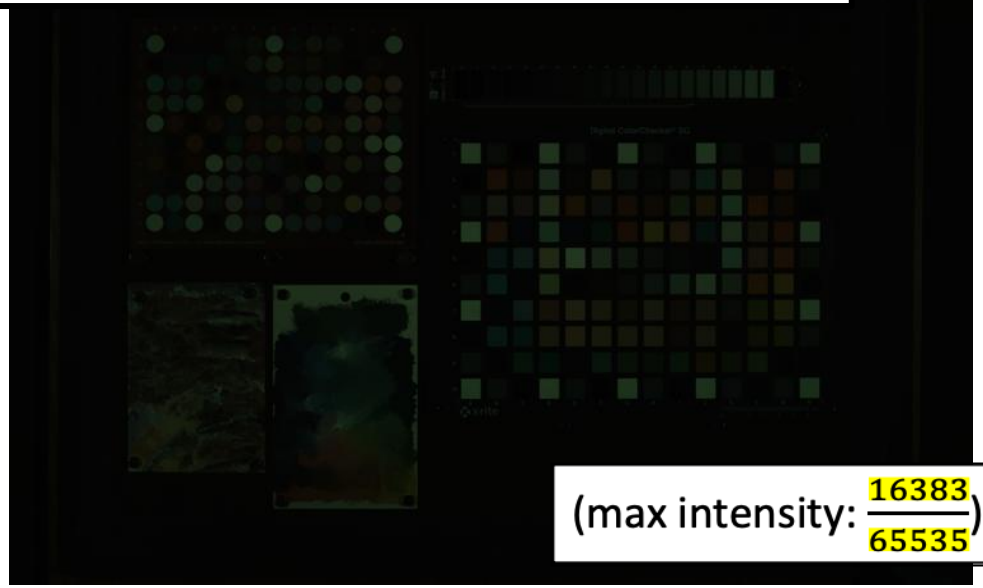
However, it is stored in an image container of 16-bit depth (65535), meaning that there are either 2- or 4- bits of unused space, causing the image to look very dark.

Therefore, we scale the RAW camera data so that the maximum value of the RAW camera data matches the maximum amount of space in the 16-bit image container. This fills and efficiently uses all the dynamic range space that we have available.

For every pixel: $scaled\ digital\ count = RAW\ digital\ count * \frac{2^{16} - 1}{2^{d} - 1}$

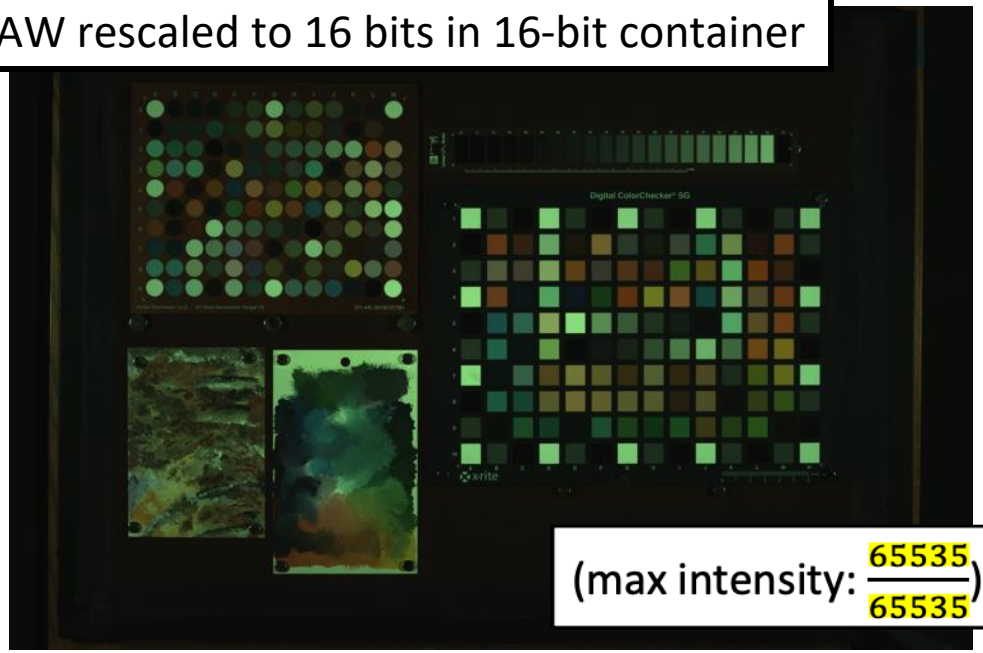
Where d is the bit depth of the RAW camera data (usually 12 or 14).

RAW as captured: 14 bits in 16-bit container



(max intensity: $\frac{16383}{65535}$)

RAW rescaled to 16 bits in 16-bit container

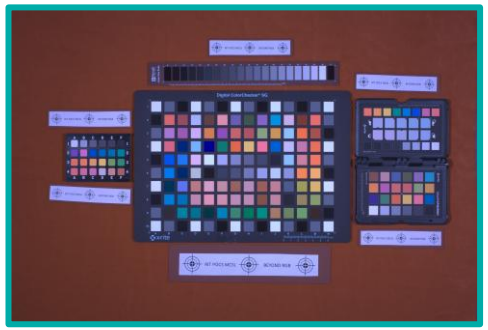


(max intensity: $\frac{65535}{65535}$)

Dark current correction:

A pixel-by-pixel operation that corrects for fixed pattern noise. These operations are performed on all flatfield, target, and object images using the corresponding dark current images.

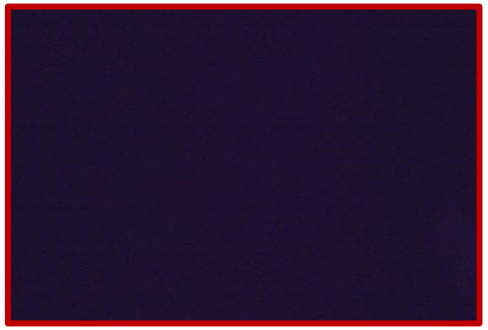
im



1. $pixel\ value_{im,n,dcc} = pixel\ value_{im,n} - pixel\ value_{dark,n}$

2. $pixel\ value_{white,n,dcc} = pixel\ value_{white,n} - pixel\ value_{dark,n}$

$dark$



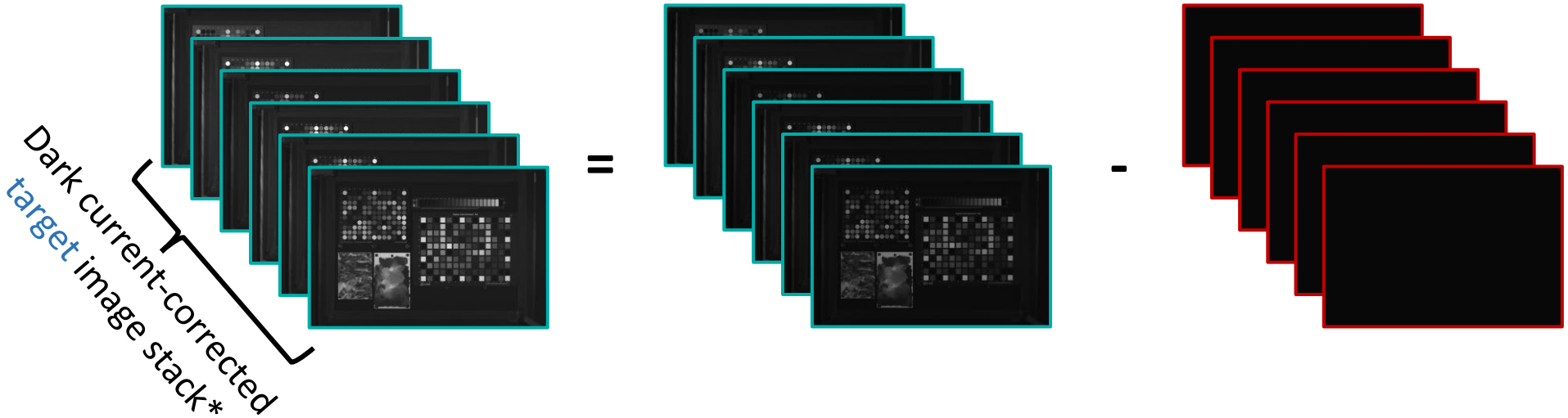
$white$

Notes:

- im refers to the image stack with scene content, including the target and object images (e.g., top images)
- $dark$ refers to the dark current image stack (e.g., middle image)
- $white$ refers to the flatfield image stack (e.g., bottom image)
- n identifies the spectral channel ($n = 1, 2, 3, 4, 5, 6$)
- dcc stands for “dark current-corrected”

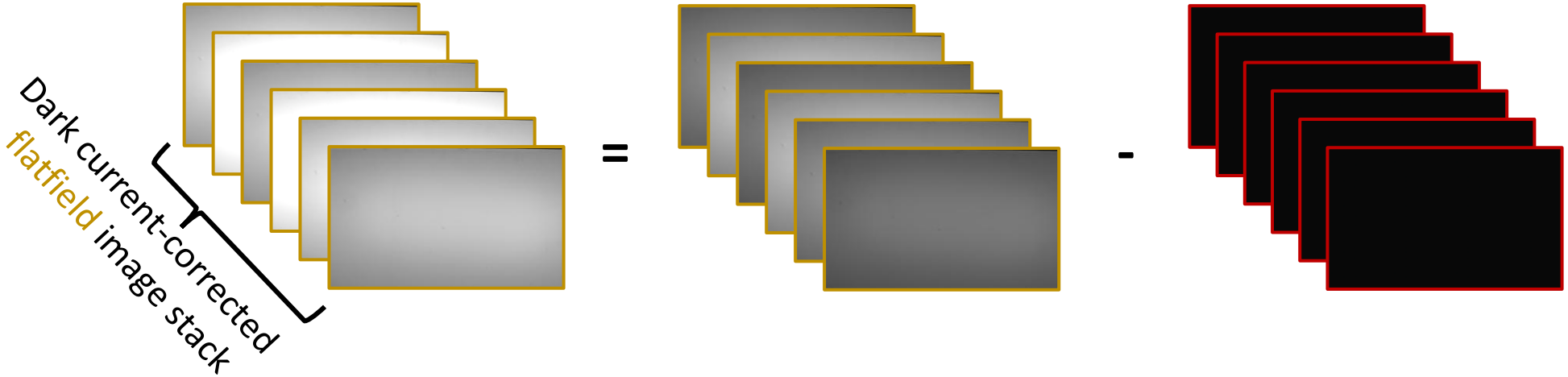
Dark current correction (visual):

1. $$pixel\ value_{im,n,dcc} = pixel\ value_{im,n} - pixel\ value_{dark,n}$$



*Also applied to the object image stack

2. $$pixel\ value_{white,n,dcc} = pixel\ value_{white,n} - pixel\ value_{dark,n}$$



Flat-fielding:

A pixel-by-pixel operation that corrects lighting non-uniformities. These operations are performed on all target, and object images using the corresponding flatfield images. This operation is performed on images that have **already been** dark current corrected.

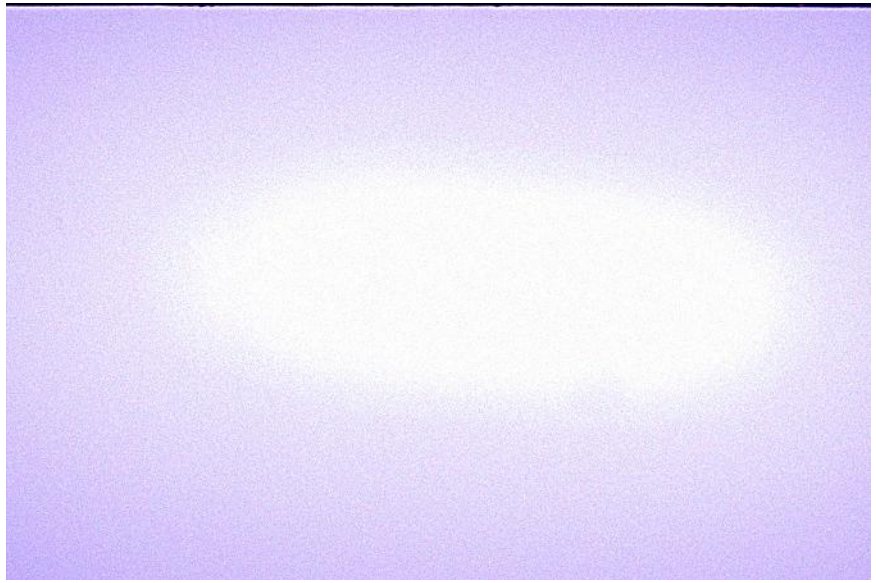
1. Apply gaussian low-pass filter (LPF) to the flatfield image stack

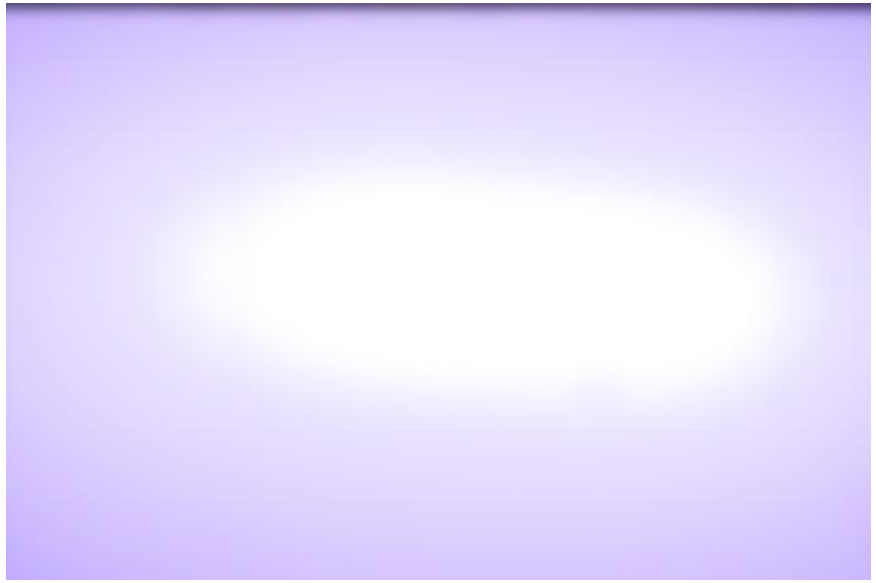
Explanation:

The flatfield image contains low spatial frequency information that describes the lighting non-uniformity across the scene, and high spatial frequency information due to image noise and board texture. Using an unfiltered flatfield image stack for flat-fielding correction corrects for lighting non-uniformity but also introduces noise into the target and object images.

Applying a low-pass filter to the flatfield image retains the low-frequency information about lighting fall-off while removing the high-frequency noise and texture. Reducing the noise that is introduced to the target and object image stacks during the flat-fielding correction.

Unfiltered flatfield image containing high-frequency noise
(Exaggerated for illustrative purposes)



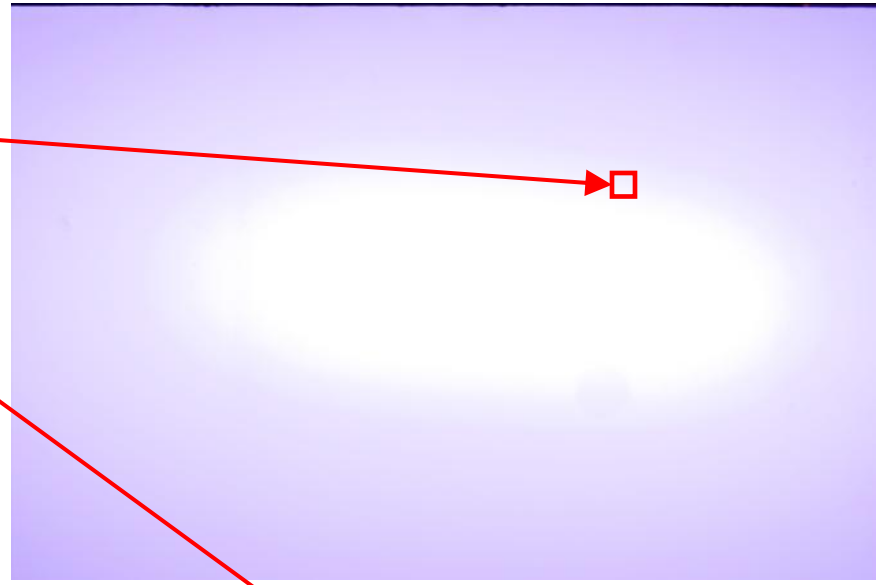


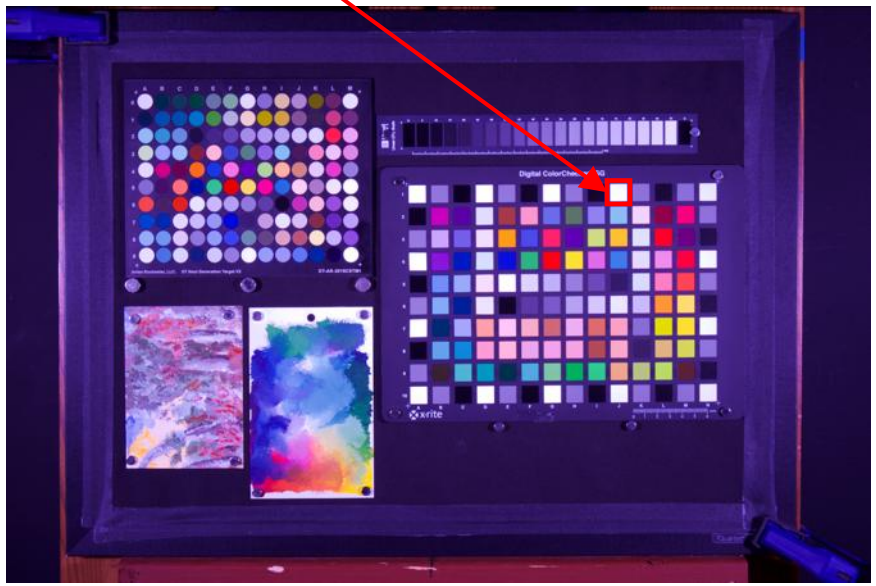
Filtered flatfield image containing only low-frequency information

Flat-fielding (cont.):

2. $$w = Y_{white\ patch,meas} * \frac{mean(white\ patch_{white,n=2,LPF})}{mean(white\ patch_{target,n=2})}$$

3. $$pixel\ value_{im,n,ff} = w * \frac{pixel\ value_{im,n,dcc}}{pixel\ value_{white,n,dcc,LPF}}$$





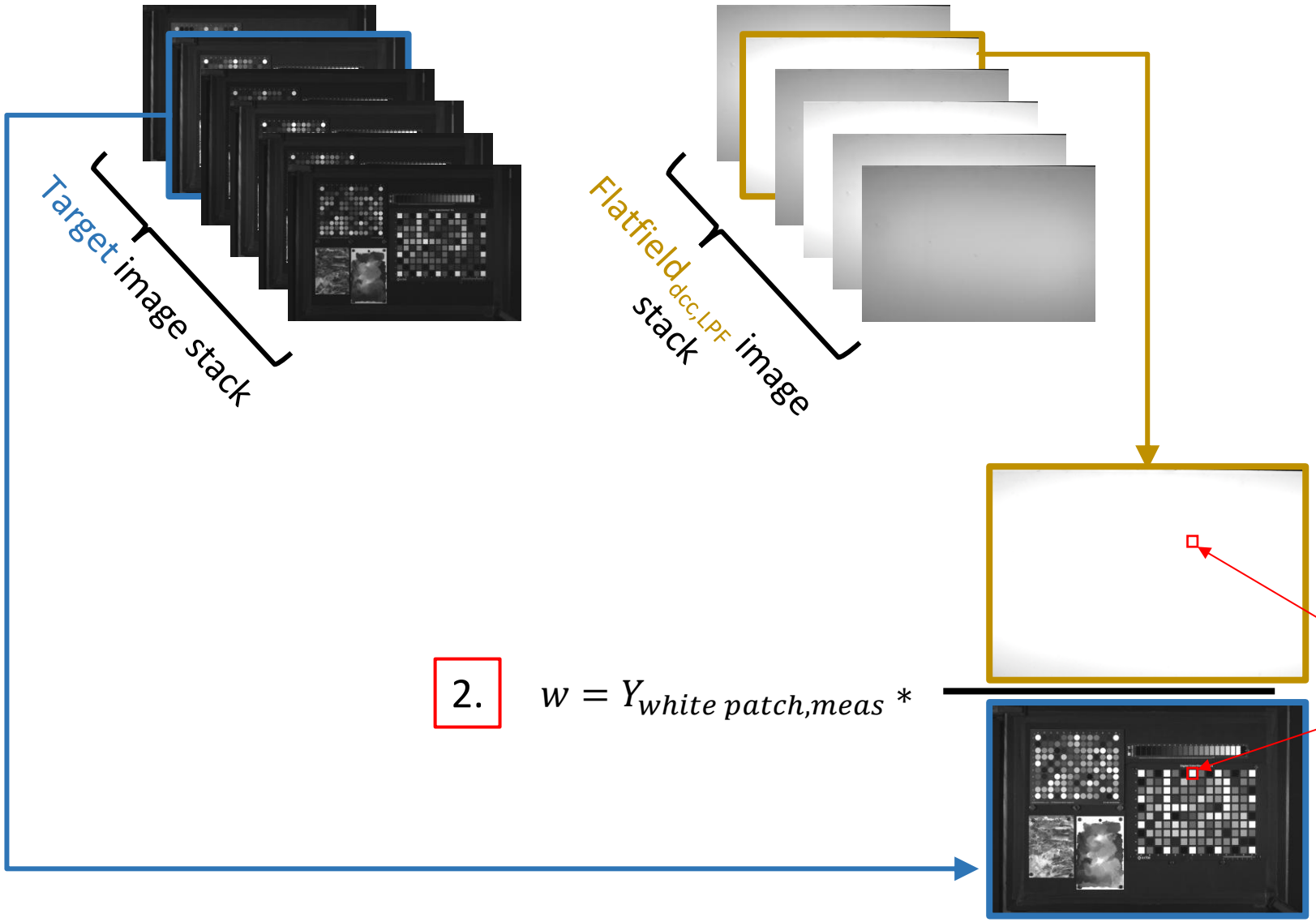
Notes:

- im refers to the image stack with content in it, including the target and object images (e.g., top images)
- $white$ refers to the flatfield image stack (e.g., bottom image)
- n identifies the spectral channel ($n = 1, 2, 3, 4, 5, 6$)
- w is a scalar multiple to account for the luminance of the flatfield – this is calculated using **only channel 2** ($n = 2$)
- dcc stands for "dark current-corrected"
- ff stands for "flat-fielded"
- LPF refers to images to which a "low-pass filter" was applied
- To check that these steps are implemented correctly: the value of the white patch after dark current correction and flatfielding should be equal to $Y_{white\ patch,meas}$

(e.g. for the CCSG target, $Y_{white\ patch,meas} = mean\ white\ patch = 0.905$)

Flat-fielding (visual) :

Flat-fielding (visual), cont.:



Pull out channel 2 from the dark current corrected target and flatfield stacks. Use **only** this channel in the division below.

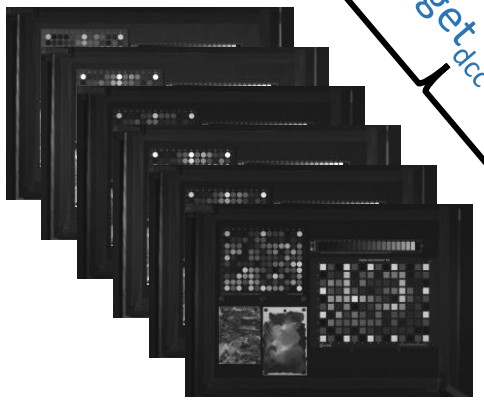
Flat-fielding (visual),cont.:

3.

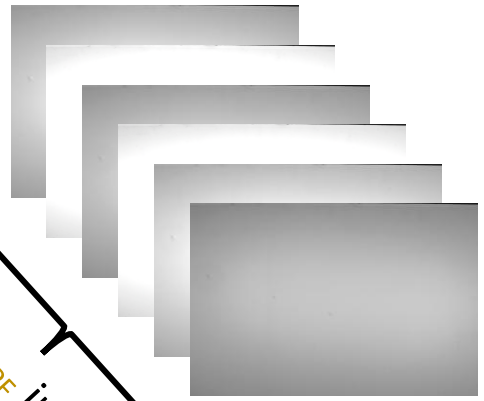
$\text{Target}_{dcc,ff}$ image stack*

*Also applied to the object_{dcc} image stack

$= w *$



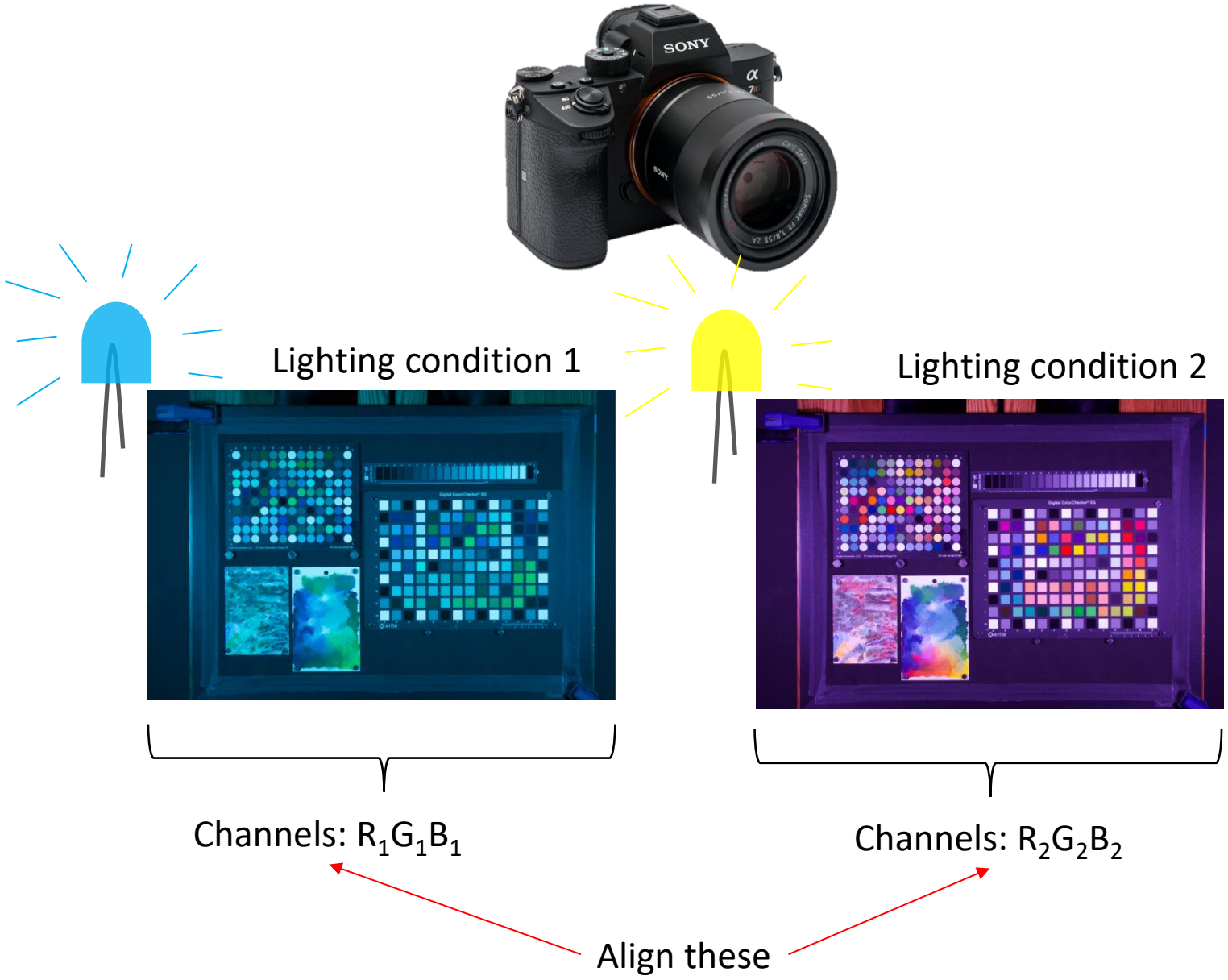
Target_{dcc} image stack*



$\text{Flatfield}_{dcc,LPF}$ image stack

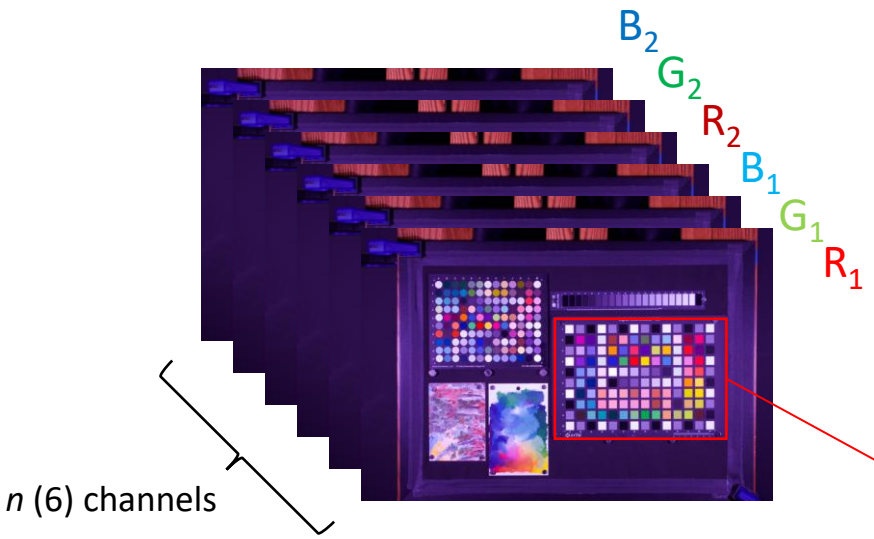
Registration:

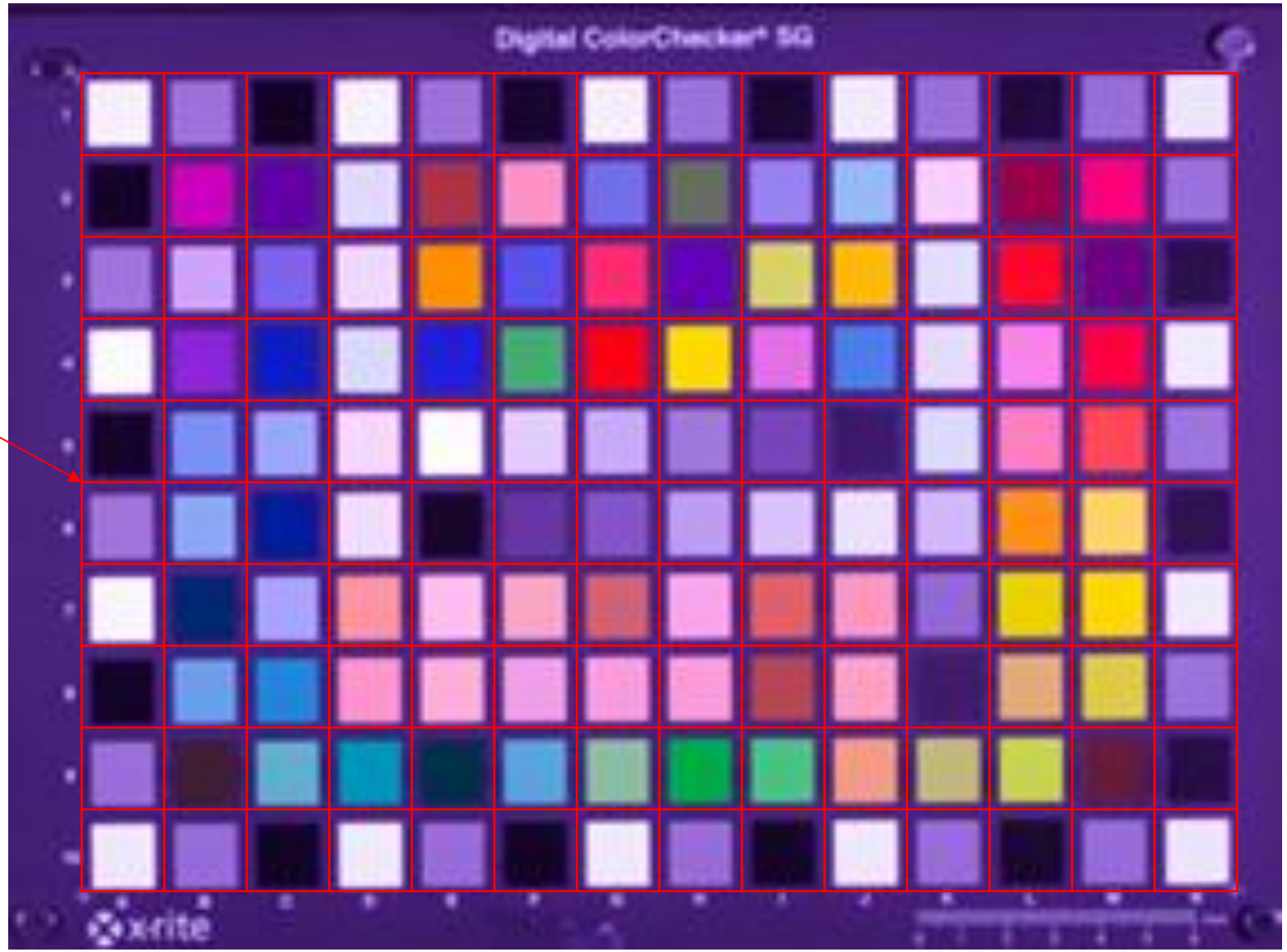
The data that the camera sensor records in each pixel will be slightly different because the light from the two different lighting conditions interacts with the optical components slightly differently. This could manifest as misalignment between the pixels of the two RGB images if you were to lay them on top of each other and compare them. Registration corrects for this misalignment by matching features in both images.



2. PROCESSING

Extract Camera Signals from Color Target Patches





Extract and average the pixel values in each of the pre-processed six channels so that we have the six $R_1G_1B_1R_2G_2B_2$ camera signal values that are associated with every single patch in the target.

The result will be a matrix of size **n x k** (number of channels x number of patches).

Colorimetric Calibration

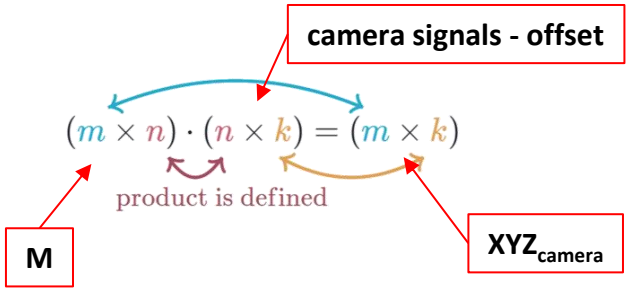
1.) Setting up the matrices to calculate the XYZ values of the patches based on the ***images*** looks like this:

n (6) channels

average values of each patch in each channel
(i.e. the digital counts data pulled from the images using the grid)

Matrix dimensions cheat sheet:

- **M**: 3 x *n* channels (6 is our default)
- **camera signals**: *n* channels x *k* target patches
- **XYZ$_{camera}$**: 3 x *k* patches



$$\boldsymbol{XYZ_{camera} = M * (camera\ signals - offset)}$$

M and *offset are* the matrices that will be optimized to minimize the color difference between the measured patch colors (reference_data.xlsx) and the imaged patch colors

which expanded looks like…

$$\begin{bmatrix} X_{patch\ 1} & X_{patch\ 2} & \cdots & X_{patch\ k} \\ Y_{patch\ 1} & Y_{patch\ 2} & \cdots & Y_{patch\ k} \\ Z_{patch\ 1} & Z_{patch\ 2} & \cdots & Z_{patch\ k} \end{bmatrix} = \begin{bmatrix} m_{1,1} & m_{1,2} & m_{1,3} & m_{1,4} & m_{1,5} & m_{1,6} \\ m_{2,1} & m_{2,2} & m_{2,3} & m_{2,4} & m_{2,5} & m_{2,6} \\ m_{3,1} & m_{3,2} & m_{3,3} & m_{3,4} & m_{3,5} & m_{3,6} \end{bmatrix} * \begin{bmatrix} camsigs_{ch\ 1,patch\ 1} - o_1 & \cdots & camsigs_{ch\ 1,patch\ k} - o_1 \\ camsigs_{ch\ 2,patch\ 1} - o_2 & \cdots & camsigs_{ch\ 2,patch\ k} - o_2 \\ camsigs_{ch\ 3,patch\ 1} - o_3 & \cdots & camsigs_{ch\ 3,patch\ k} - o_3 \\ camsigs_{ch\ 4,patch\ 1} - o_4 & \cdots & camsigs_{ch\ 4,patch\ k} - o_4 \\ camsigs_{ch\ 5,patch\ 1} - o_5 & \cdots & camsigs_{ch\ 5,patch\ k} - o_5 \\ camsigs_{ch\ 6,patch\ 1} - o_6 & & camsigs_{ch\ 6,patch\ k} - o_6 \end{bmatrix}$$

Colorimetric Calibration (cont.)

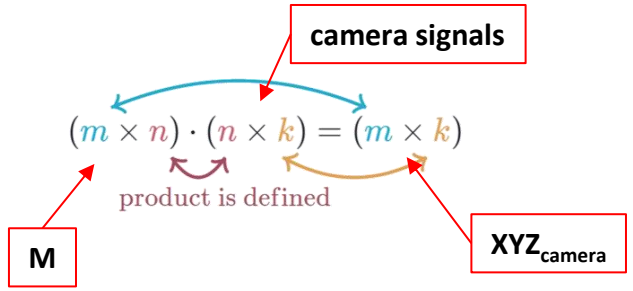
1a.) Starting the optimization process requires providing starting values with which to populate *M* and *offset*:

$$M_{initial} = \begin{bmatrix} 0.1 & 0.1 & 0.25 & 0.5 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.25 & 0.1 & 1 & 0.1 \\ 0.1 & 0.1 & 0.25 & 0.1 & 0.1 & 0.5 \end{bmatrix}$$

$$offset_{initial} = [\,0.01 \quad 0.01 \quad 0.01 \quad 0.01 \quad 0.01 \quad 0.01\,]'$$

Matrix dimensions cheat sheet:

- **M**: 3 x *n* channels (6 is our default)
- **camera signals**: *n* channels x *k* target patches
- **XYZ$_{camera}$**: 3 x *k* patches



$$\boldsymbol{XYZ_{camera} = M * (camera\ signals - offset)}$$

which expanded looks like…

$$\begin{bmatrix} X_{patch\,1} & X_{patch\,2} & \cdots & X_{patch\,k} \\ Y_{patch\,1} & Y_{patch\,2} & \cdots & Y_{patch\,k} \\ Z_{patch\,1} & Z_{patch\,2} & \cdots & Z_{patch\,k} \end{bmatrix} = \begin{bmatrix} m_{1,1} & m_{1,2} & m_{1,3} & m_{1,4} & m_{1,5} & m_{1,6} \\ m_{2,1} & m_{2,2} & m_{2,3} & m_{2,4} & m_{2,5} & m_{2,6} \\ m_{3,1} & m_{3,2} & m_{3,3} & m_{3,4} & m_{3,5} & m_{3,6} \end{bmatrix} * \begin{bmatrix} camsigs_{ch\,1,patch\,1} - o_1 & & camsigs_{ch\,1,patch\,k} - o_1 \\ camsigs_{ch\,2,\,patch\,1} - o_2 & \cdots & camsigs_{ch\,2,\,patch\,k} - o_2 \\ camsigs_{ch\,3,\,patch\,1} - o_3 & \cdots & camsigs_{ch\,3,\,patch\,k} - o_3 \\ camsigs_{ch\,4,\,patch\,1} - o_4 & \cdots & camsigs_{ch\,4,\,patch\,k} - o_4 \\ camsigs_{ch\,5,\,patch\,1} - o_5 & \cdots & camsigs_{ch\,5,\,patch\,k} - o_5 \\ camsigs_{ch\,6,\,patch\,1} - o_6 & & camsigs_{ch\,6,\,patch\,k} - o_6 \end{bmatrix}$$

Colorimetric Calibration (cont.)

2.) Iteratively change the coefficients in *M* until the average color difference (ΔE_{00}) between the colors of all of the patches as captured by the camera (XYZ_{camera} ⍰ $L^*a^*b^*_{camera}$) and the ground truth measurements ($L^*a^*b^*_{reference}$) is minimized

OPTIMIZATION

CAMERA

1.) Plug some values into these two matrices

2.) Calculate this

$$XYZ_{camera} = M * (camera\ signals - offset)$$

(XYZ to L*a*b* functions)

matrix transformation noise

3.) Then calculate this: $L^*a^*b^*_{camera}$ and N

REALITY

Reference target data (reference_data.xlsx)

OR

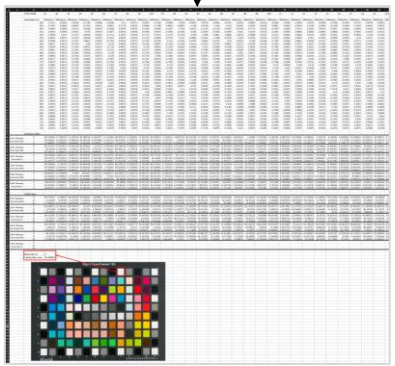
Measurements provided by user

Gives

$XYZ_{reference}$

and

$L^*a^*b^*_{reference}$



CIE ΔE_{00} formula

Minimum **mean** ΔE_{00} + $0.1N$?

no → Try new coefficients in *M* and *offset* and repeat

yes → Yay, your matrix is optimized! You're done!

Calculating XYZ (tristimulus values) of a reflectance sample:

$$X = k \int_{\lambda} S_{\lambda} R_{\lambda} \bar{x}_{\lambda} d\lambda$$

$$Y = k \int_{\lambda} S_{\lambda} R_{\lambda} \bar{y}_{\lambda} d\lambda$$

$$Z = k \int_{\lambda} S_{\lambda} R_{\lambda} \bar{z}_{\lambda} d\lambda$$

where

$$k = \frac{100}{\int_{\lambda} S_{\lambda} \bar{y}_{\lambda} d\lambda}$$

OR EQUIVALENTLY…

$$X = k \sum_{\lambda} S_{\lambda} R_{\lambda} \bar{x}_{\lambda} \Delta\lambda$$

$$Y = k \sum_{\lambda} S_{\lambda} R_{\lambda} \bar{y}_{\lambda} \Delta\lambda$$

$$Z = k \sum_{\lambda} S_{\lambda} R_{\lambda} \bar{z}_{\lambda} \Delta\lambda$$

where

$$k = \frac{100}{\sum_{\lambda} S_{\lambda} \bar{y}_{\lambda} \Delta\lambda}$$

S_{λ} is the spectral power distribution of the illuminating light source.

R_{λ} is the spectral reflectance of the reflectance sample (patch on target).

$\bar{x}_{,\lambda}$, $\bar{y}_{\lambda}$, and $\bar{z}_{\lambda}$ are called Color Matching Functions (CMFs), and together, they make up a "Standard Observer." There are two different Standard Observers, differentiated by dates (1931 and 1964).

$\Delta\lambda$ is the sampling increment (*i.e.* each of the above are provided over the range of 380 nm to 780 nm, with a sampling increment of 10 nm between each data point)

See reference_data.xlsx for all of this data + a worked example calculation

XYZ to Lab:

See reference_data.xlsx

$$L^* = 116 * f\left(\frac{Y}{Y_n}\right) - 16$$

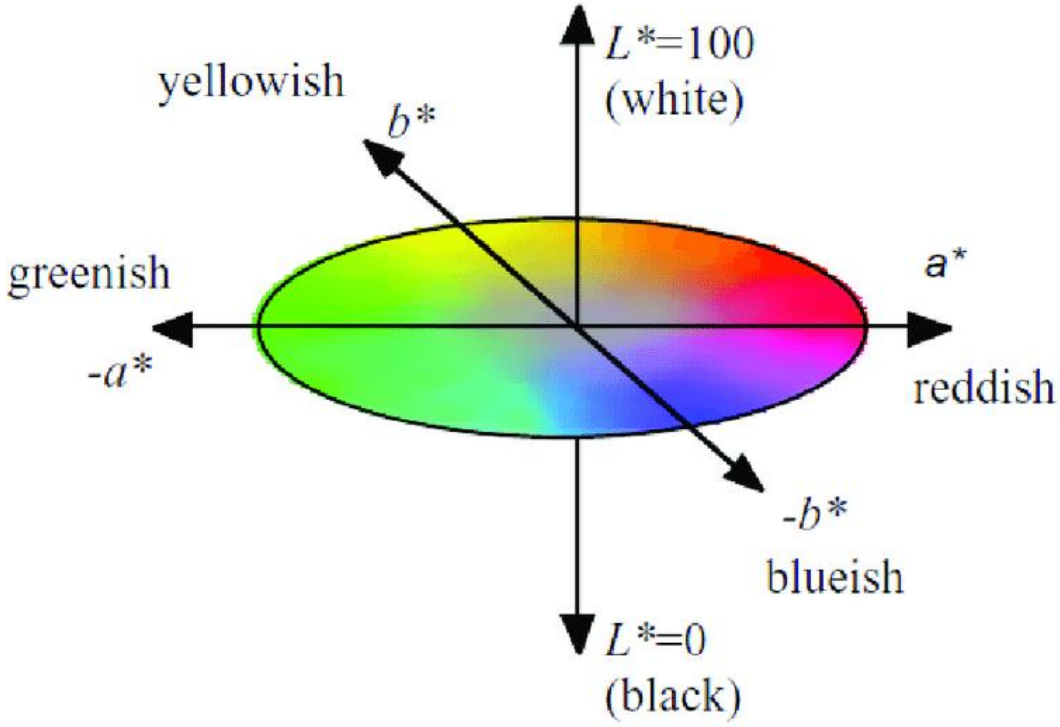
$$a^* = 500 * \left[f\left(\frac{X}{X_n}\right) - f\left(\frac{Y}{Y_n}\right)\right]$$

$$b^* = 200 * \left[f\left(\frac{Y}{Y_n}\right) - f\left(\frac{Z}{Z_n}\right)\right]$$

where

$$f(x) = \begin{cases} x^{1/3}, & x > \frac{216}{24389} \\ \dfrac{\frac{24389}{27}x + 16}{116}, & otherwise \end{cases}$$

and X_n, Y_n, and Z_n are the XYZ values of the illuminant's white point



White Points						
Observer	**1931 (2˚)**			**1964 (10˚)**		
Illuminant	**A**	**D50**	**D65**	**A**	**D50**	**D65**
X_n	109.846607	96.4211994	95.0428545	111.142041	96.7206275	94.8096677
Y_n	100	100	100	100	100	100
Z_n	35.58228	82.5188285	108.890037	35.1997832	81.4280151	107.305136

CIEDE2000 Color Difference Calculation:

- **Input**: a pair of colors defined by their L*a*b* values
- **Output**: a number that quantifies how different they look to the average person
- ***Why do we have to do this?***
  - This is how we quantify the differences between the objective (measured) values of the color target patches and the values of the patches as captured by the camera
  - Then we'll do an optimization to minimize these differences, so that the imaged values are as close to the objective values as possible
    - This optimization is the "calibration" that is the meat of the processing stage

Notes:

$$k_L = 1$$
$$k_C = 1$$
$$k_H = 1$$

$$C^* = \sqrt{a^{*2} + b^{*2}}$$

$$\Delta E^*_{00} = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \frac{\Delta C'}{k_C S_C}\frac{\Delta H'}{k_H S_H}}$$

$$\Delta L' = L^*_2 - L^*_1$$

$$\bar{L} = \frac{L^*_1 + L^*_2}{2} \quad \bar{C} = \frac{C^*_1 + C^*_2}{2}$$

$$a'_1 = a^*_1 + \frac{a^*_1}{2}\left(1 - \sqrt{\frac{\bar{C}^7}{\bar{C}^7 + 25^7}}\right) \quad a'_2 = a^*_2 + \frac{a^*_2}{2}\left(1 - \sqrt{\frac{\bar{C}^7}{\bar{C}^7 + 25^7}}\right)$$

$$\bar{C}' = \frac{C'_1 + C'_2}{2} \text{ and } \Delta C' = C'_2 - C'_1 \quad \text{where } C'_1 = \sqrt{a'^2_1 + b^{*2}_1} \quad C'_2 = \sqrt{a'^2_2 + b^{*2}_2}$$

$$h'_1 = \text{atan2}(b^*_1, a'_1) \mod 360^\circ, \quad h'_2 = \text{atan2}(b^*_2, a'_2) \mod 360^\circ$$

$$\Delta h' = \begin{cases} h'_2 - h'_1 & |h'_1 - h'_2| \le 180^\circ \\ h'_2 - h'_1 + 360^\circ & |h'_1 - h'_2| > 180^\circ, h'_2 \le h'_1 \\ h'_2 - h'_1 - 360^\circ & |h'_1 - h'_2| > 180^\circ, h'_2 > h'_1 \end{cases}$$

$$\Delta H' = 2\sqrt{C'_1 C'_2}\sin(\Delta h'/2), \quad \bar{H}' = \begin{cases} (h'_1 + h'_2 + 360^\circ)/2 & |h'_1 - h'_2| > 180^\circ \\ (h'_1 + h'_2)/2 & |h'_1 - h'_2| \le 180^\circ \end{cases}$$

$$T = 1 - 0.17\cos(\bar{H}' - 30^\circ) + 0.24\cos(2\bar{H}') + 0.32\cos(3\bar{H}' + 6^\circ) - 0.20\cos(4\bar{H}' - 63^\circ)$$

$$S_L = 1 + \frac{0.015\left(\bar{L} - 50\right)^2}{\sqrt{20 + \left(\bar{L} - 50\right)^2}} \quad S_C = 1 + 0.045\bar{C}' \quad S_H = 1 + 0.015\bar{C}'T$$

$$R_T = -2\sqrt{\frac{\bar{C}'^7}{\bar{C}'^7 + 25^7}}\sin\left[60^\circ \cdot \exp\left(-\left[\frac{\bar{H}' - 275^\circ}{25^\circ}\right]^2\right)\right]$$

Matrix Transformation Noise Formula:

- **Input**: 3x6 color transformation matrix
- **Output**: a number that quantifies how much the matrix amplifies visual noise in the image
- ***Why do we have to do this?***
  - Every calculation propagates noise. The transformation matrix can be a source of very large noise amplification.
  - This noise can cause visible artifacts in the image and makes it very hard to reproduce results.
  - Including the noise parameter as part of the cost function in the optimization penalizes matrices that would significantly increase noise.

Notes: The noise parameter only needs to be calculated for a middle gray patch. Therefore, the value of A remains constant. For a middle gray, A is:

$$A = \frac{{Y_{gray}}^{-2/3}}{{Y_{white}}^{1/3}} = \frac{{18.4}^{-2/3}}{{100}^{1/3}}$$

Remember from slide 17, the colorimetric transformation matrix takes the form:

$$M = \begin{bmatrix} m_{1,1} & m_{1,2} & m_{1,3} & m_{1,4} & m_{1,5} & m_{1,6} \\ m_{2,1} & m_{2,2} & m_{2,3} & m_{2,4} & m_{2,5} & m_{2,6} \\ m_{3,1} & m_{3,2} & m_{3,3} & m_{3,4} & m_{3,5} & m_{3,6} \end{bmatrix}$$

To calculate the noise parameter (N):

1. Rescale the matrix so that the sum of each row is 1.

$$M' = \begin{bmatrix} k_1 & 0 & 0 \\ 0 & k_2 & 0 \\ 0 & 0 & k_3 \end{bmatrix} \begin{bmatrix} m_{1,1} & m_{1,2} & m_{1,3} & m_{1,4} & m_{1,5} & m_{1,6} \\ m_{2,1} & m_{2,2} & m_{2,3} & m_{2,4} & m_{2,5} & m_{2,6} \\ m_{3,1} & m_{3,2} & m_{3,3} & m_{3,4} & m_{3,5} & m_{3,6} \end{bmatrix}$$

Where: $\sum_{n=1}^{6} k_1 m_{1,n} = \sum_{n=1}^{6} k_2 m_{2,n} = \sum_{n=1}^{6} k_3 m_{3,n} = 1$

2. $$\Delta {L^*}_N{}^2 = \left(\frac{116A}{3}\right)^2 \sum_{n=1}^{6} {m'}_{2,n}{}^2$$

$$\Delta {a^*}_N{}^2 = \left(\frac{500A}{3}\right)^2 \sum_{n=1}^{6} \left({m'}_{1,n} - {m'}_{2,n}\right)^2$$

$$\Delta {b^*}_N{}^2 = \left(\frac{200A}{3}\right)^2 \sum_{n=1}^{6} \left({m'}_{2,n} - {m'}_{3,n}\right)^2$$

Use the rescaled matrix (M') to calculate these parameters

3. $$N = \sqrt{\Delta {L^*}_N{}^2 + \Delta {a^*}_N{}^2 + \Delta {b^*}_N{}^2}$$

Final noise parameter to use in optimization

Spectral Reflectance Calibration

1.) Setting up the matrices to calculate the spectral reflectance of the patches based on the ***images*** looks like this:

n (6) channels

average values of each patch in each channel
(i.e. the digital counts data pulled from the images using the grid)

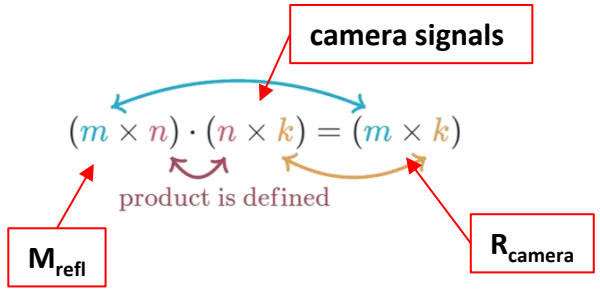
$$\boldsymbol{R_{camera} = M_{refl} * camera\ signals}$$

which expanded looks like…

$$\begin{bmatrix} R_{\lambda_1,1} & R_{\lambda_1,2} & \cdots & R_{\lambda_1,k} \\ R_{\lambda_2,1} & R_{\lambda_2,2} & \cdots & R_{\lambda_2,k} \\ \vdots & \vdots & \cdots & \vdots \\ R_{\lambda_{36},1} & R_{\lambda_{36},2} & \cdots & R_{\lambda_{36},k} \end{bmatrix} = \boxed{\begin{bmatrix} m_{1,1} & m_{1,2} & m_{1,3} & m_{1,4} & m_{1,5} & m_{1,6} \\ m_{2,1} & m_{2,2} & m_{2,3} & m_{2,4} & m_{2,5} & m_{2,6} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ m_{36,1} & m_{36,2} & m_{36,3} & m_{36,4} & m_{36,5} & m_{36,6} \end{bmatrix}} * \begin{bmatrix} camsigs_{ch\ 1, patch\ 1} & & camsigs_{ch\ 1, patch\ k} \\ camsigs_{ch\ 2, patch\ 1} & \cdots & camsigs_{ch\ 2, patch\ k} \\ camsigs_{ch\ 3, patch\ 1} & \vdots & camsigs_{ch\ 3, patch\ k} \\ camsigs_{ch\ 4, patch\ 1} & \vdots & camsigs_{ch\ 4, patch\ k} \\ camsigs_{ch\ 5, patch\ 1} & \cdots & camsigs_{ch\ 5, patch\ k} \\ camsigs_{ch\ 6, patch\ 1} & & camsigs_{ch\ 6, patch\ k} \end{bmatrix}$$

Matrix dimensions cheat sheet:

- **M**: 36 x *n* channels (6 is our default)
- **camera signals**: *n* channels x *k* target patches
- $\mathbf{R_{camera}}$: 36 x *k* patches



M_{refl} is the matrix that will be optimized to minimize the RMS error between the measured patch spectral reflectances (reference_data.xlsx) and the imaged patch spectral reflectances

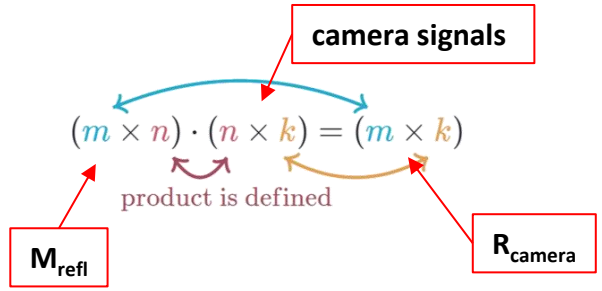
Spectral Reflectance Calibration (cont.)

1a.) Starting the optimization process requires providing starting values with which to populate M_{refl}:

$$M_{refl,initial} = \underbrace{R_{reference}}_{\text{From reference_data.xlsx}} * pseudoinverse(camera\ signals)$$

Matrix dimensions cheat sheet:

- **M**: 36 x *n* channels (6 is our default)
- **camera signals**: *n* channels x *k* target patches
- $\mathbf{R_{camera}}$: 36 x *k* patches



$$\boldsymbol{R_{camera} = M_{refl} * camera\ signals}$$

which expanded looks like…

$$\begin{bmatrix} R_{\lambda_1,1} & R_{\lambda_1,2} & \cdots & R_{\lambda_1,k} \\ R_{\lambda_2,1} & R_{\lambda_2,2} & \cdots & R_{\lambda_2,k} \\ \vdots & \vdots & \cdots & \vdots \\ R_{\lambda_{36},1} & R_{\lambda_{36},2} & \cdots & R_{\lambda_{36},k} \end{bmatrix} = \begin{bmatrix} m_{1,1} & m_{1,2} & m_{1,3} & m_{1,4} & m_{1,5} & m_{1,6} \\ m_{2,1} & m_{2,2} & m_{2,3} & m_{2,4} & m_{2,5} & m_{2,6} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ m_{36,1} & m_{36,2} & m_{36,3} & m_{36,4} & m_{36,5} & m_{36,6} \end{bmatrix} * \begin{bmatrix} camsigs_{ch\ 1, patch\ 1} & & camsigs_{ch\ 1, patch\ k} \\ camsigs_{ch\ 2, patch\ 1} & \cdots & camsigs_{ch\ 2, patch\ k} \\ camsigs_{ch\ 3, patch\ 1} & \vdots & camsigs_{ch\ 3, patch\ k} \\ camsigs_{ch\ 4, patch\ 1} & \vdots & camsigs_{ch\ 4, patch\ k} \\ camsigs_{ch\ 5, patch\ 1} & \cdots & camsigs_{ch\ 5, patch\ k} \\ camsigs_{ch\ 6, patch\ 1} & & camsigs_{ch\ 6, patch\ k} \end{bmatrix}$$

Spectral Reflectance Calibration (cont.)

2.) Iteratively change the coefficients in M_{refl} until the weighted error term Z, which sums RMS/SS error differences between the reference reflectance spectra and the reflectance spectra estimated from the camera data, is minimized

OPTIMIZATION

CAMERA

1.) Plug some values into this matrix

2.) Calculate this

$$R_{camera} = M_{refl} * (camera\ signals)$$

and find

$R_{camera,global\ max}$ & $R_{camera,global\ min}$

(at every wavelength (λ))

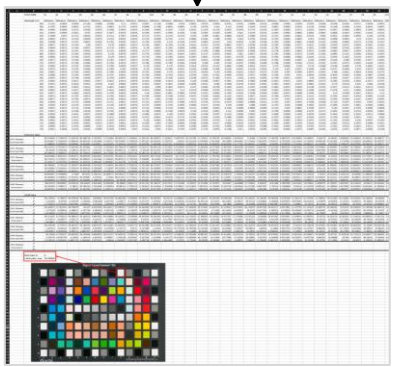
3.) Calculate these error terms

REALITY

Reference target data (reference_data.xlsx)

OR

Measurements provided by user



Gives

$R_{reference}$

find

$R_{reference,global\ max}$ & $R_{reference,global\ min}$

(at every wavelength (λ))

$$Z = e_1 + 10e_2 + 50e_3$$

$$e_1 = \sqrt{\sum_{\lambda}\sum_{p}\frac{1}{p}(R_{reference,p,\lambda} - R_{camera,p,\lambda})^2}$$

$$e_2 = \sum_{\lambda}(R_{camera,global\ max,\lambda} - R_{reference,global\ max,\lambda})^2$$

$$e_3 = \sum_{\lambda}(R_{camera,global\ min,\lambda} - R_{reference,global\ min,\lambda})^2$$

Minimum ***Z***?

no → Try new coefficients in M_{refl} and repeat

yes → Yay, your matrix is optimized! You're done!

Optional: VERIFICATION

Verification of Colorimetric Calibration

VERIFICATION

CAMERA

1.) Plug the optimized *M* and *offset* matrices in

2.) Calculate this

$$XYZ_{camera} = M * (camera\ signals - offset)$$

(XYZ to L*a*b* functions)

1a.) Plug the camera signals from the second target in the image in here

3.) Then calculate this: $L^*a^*b^*_{camera}$

REALITY

Reference target data (reference_data.xlsx)

OR

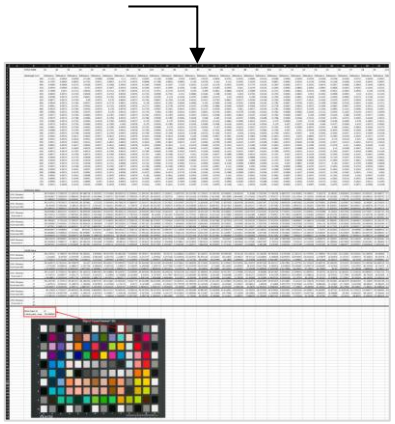
Measurements provided by user

Gives

$XYZ_{reference}$

and

$L^*a^*b^*_{reference}$



4.) Then this: CIE ΔE_{00} formula

Save XYZ_{camera}, $L^*a^*b^*_{camera}$, and the ΔE_{00} values for exporting/reporting in the viewer with the calibration results

Verification of Spectral Reflectance Calibration

VERIFICATION

CAMERA

1.) Plug the optimized M_{refl} in here

2.) Calculate this

$$R_{camera} = M_{refl} * (camera\ signals)$$

1a.) Plug the camera signals from the second target in the image in here

3.) Then calculate RMSE:

$$RMSE = \sqrt{\sum_{i=1}^{N} \frac{\left(R_{camera,i} - R_{reference,i}\right)^2}{N}}$$

REALITY

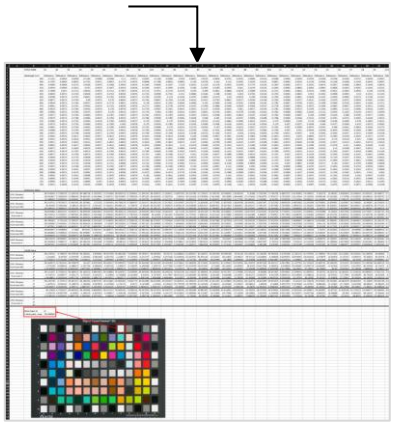
Reference target data (reference_data.xlsx)

OR

Measurements provided by user

Gives

$R_{reference}$



Save R_{camera} and RMSE for exporting/reporting in the viewer with the calibration results

4. RENDERING & OUTPUT

Computing the Color-Calibrated Image

1.) Apply the **optimized** *M* and *offset* to every pixel of every channel to compute the preliminary XYZ version of the final image:

$$\boldsymbol{XYZ_{camera} = M * (camera\ signals - offset)}$$

which expands to…

$$\begin{bmatrix} X_{pixel\ 1} & X_{pixel\ 2} & \cdots & X_{pixel\ k} \\ Y_{pixel\ 1} & Y_{pixel\ 2} & \cdots & Y_{pixel\ k} \\ Z_{ixel\ 1} & Z_{pixel\ 2} & \cdots & Z_{pixel\ k} \end{bmatrix} = \begin{bmatrix} m_{1,1} & m_{1,2} & m_{1,3} & m_{1,4} & m_{1,5} & m_{1,6} \\ m_{2,1} & m_{2,2} & m_{2,3} & m_{2,4} & m_{2,5} & m_{2,6} \\ m_{3,1} & m_{3,2} & m_{3,3} & m_{3,4} & m_{3,5} & m_{3,6} \end{bmatrix} * \begin{bmatrix} cam\ sig_{ch\ 1, pixel\ 1} - o_1 & & cam\ sig_{ch\ 1, pixel\ k} - o_1 \\ cam\ sig_{ch\ 2, pixel\ 1} - o_2 & \cdots & cam\ sig_{ch\ 2, pixel\ k} - o_2 \\ cam\ sig_{ch\ 3, pixel\ 1} - o_3 & \vdots & cam\ sig_{ch\ 3, pixel\ k} - o_3 \\ cam\ sig_{ch\ 4, pixel\ 1} - o_4 & \vdots & cam\ sig_{ch\ 4, pixel\ k} - o_4 \\ cam\ sig_{ch\ 5, pixel\ 1} - o_5 & \cdots & cam\ sig_{ch\ 5, pixel\ k} - o_5 \\ cam\ sig_{ch\ 6, patch\ 1} - o_6 & & cam\ sig_{ch\ 6, pixel\ k} - o_6 \end{bmatrix}$$

…and will give a 3-by-(image rows x image cols) matrix of the XYZ values of every pixel.

2.) Convert to linear ProPhoto color space using the XYZ to RGB matrix (M_{PP}) provided below:

$$\boldsymbol{RGB_{PP,\ linear} = M * XYZ_{camera}}$$

$$M = \begin{bmatrix} 1.3459433 & -0.2556075 & -0.0511118 \\ -0.5445989 & 1.5081673 & 0.0205351 \\ 0.0000000 & 0.0000000 & 1.2118128 \end{bmatrix}$$

Computing the Color-Calibrated Image

3.) Clip the linear RGB values between 0 and 1:

$$\boldsymbol{RGB_{PP,\,linear,\,clipped} = f(RGB_{PP,\,linear})}$$

$$f(x) = \begin{cases} 0\,, & x < 0 \\ x\,, & 0 \le x \le 1 \\ 1\,, & x > 1 \end{cases}$$

4.) Apply gamma:

$$\boldsymbol{RGB_{PP,final} = f(RGB_{PP,\,linear,clipped})}$$

$$f(x) = \begin{cases} x^{1/1.8}\,, & x \ge 0.001953125 \\ x * 16\,, & otherwise \end{cases}$$

Going from $RGB_{ProPhoto}$ to XYZ to RGB_{sRGB}

1.) Re-linearize the RGB_{PP} values by applying the inverse gamma:

$$\boldsymbol{RGB_{PP,linear} = f(RGB_{PP,final})}$$

$$f(x) = \begin{cases} x^{1.8}, & x \geq 16*0.001953125 \\ \dfrac{x}{16}, & otherwise \end{cases}$$

2.) Convert from linear ProPhoto RGB values back to XYZ values by applying the inverse transform (M_{PP}^{-1}) provided below:

$$\boldsymbol{XYZ = M^{-1} * RGB_{PP,\,linear}}$$

$$M^{-1} = \begin{bmatrix} 0.7976749 & 0.1351917 & 0.0313534 \\ 0.2880402 & 0.7118741 & 0.0000857 \\ 0.0000000 & 0.0000000 & 0.8252100 \end{bmatrix}$$

3.) Convert to linear sRGB color space using the XYZ to RGB matrix (M_{sRGB}) provided below:

$$\boldsymbol{RGB_{sRGB,\,linear} = M_{sRGB} * XYZ}$$

$$M_{sRGB} = \begin{bmatrix} 3.1338561 & -1.6168667 & -0.4906146 \\ -0.9787684 & 1.9161415 & 0.0334540 \\ 0.0719453 & -0.2289914 & 1.4052427 \end{bmatrix}$$

4.) Clip the linear sRGB values between 0 and 1:

$$\boldsymbol{RGB_{sRGB,\,linear,\,clipped} = f(RGB_{sRGB,\,linear})}$$

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

5.) Apply the sRGB gamma:

$$\boldsymbol{RGB_{sRGB,final} = f(RGB_{sRGB,\,linear,clipped})}$$

$$f(x) = \begin{cases} 1.055 * x^{1/2.4} - 0.055, & x > 0.0031308 \\ x * 12.92, & otherwise \end{cases}$$

SPECTRAL PICKING

Computing Estimated Reflectance Spectra in the Spectral Picker:

1.) Average the digital counts for each of the 6 channels from all the pixels in the selected region (this will give a 6-by-1 matrix of camera signal values – see below).

2.) Apply the **optimized** M_{refl} to the averaged camera signals:

$$\boldsymbol{R_{camera} = M_{refl} * camera\ signals}$$

which expands to…

$$\begin{bmatrix} R_{camera\,\lambda_1} \\ R_{camera\,\lambda_2} \\ \vdots \\ R_{camera\,\lambda_{36}} \end{bmatrix} = \begin{bmatrix} m_{1,1} & m_{1,2} & m_{1,3} & m_{1,4} & m_{1,5} & m_{1,6} \\ m_{2,1} & m_{2,2} & m_{2,3} & m_{2,4} & m_{2,5} & m_{2,6} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ m_{36,1} & m_{36,2} & m_{36,3} & m_{36,4} & m_{36,5} & m_{36,6} \end{bmatrix} * \begin{bmatrix} cam\ sig_{ch\ 1\ average} \\ cam\ sig_{ch\ 2\ average} \\ cam\ sig_{ch\ 3\ average} \\ cam\ sig_{ch\ 4\ average} \\ cam\ sig_{ch\ 5,\ average} \\ cam\ sig_{ch\ 6\ average} \end{bmatrix}$$

…and will give a 36-by-1 matrix of the reflectance spectrum of the averaged pixels.